

PROBABILITY THEORY AND RANDOM PROCESS : ECECC303

A. SYLLABUS

Course No.	Title of the Course	Course Structure	Pre-Requisite
ECECC303	Probability Theory and Random Process	L-T-P 3-1-0	None
COURSE OUTCOMES (CO):			
completing the course, the students should be able to: Understand the concepts of probability theory and random variables for the analysis of random phenomena. Evaluate and apply moments & characteristic functions and understand the concept of inequalities. Characterize probability models and function of random variables based on single & multiple random variables. Understand the concept of random processes and its characteristics. Evaluate and analyse the Spectral and temporal characteristics of the response of LTI systems to a random process.			
Unit No.	Topics		
Unit 1	Probability Theory: Introduction to Random Variables, PDF, CDF, Continuous/Discrete/Mixed type random variables. Normal, Exponential, chi-square, Rayleigh, Nakagami-m, uniform distributions, etc. Bernoulli, Binomial, Poisson distributions. Expectation, Variance, Moments, Characteristic functions.		
Unit 2	Joint Random variables and random vectors. Independent, uncorrelated random variables. Joint Expectation, co-variance, Joint Moments, joint characteristic functions, conditional distributions, conditional expectation. Multidimensional Gaussian law		
Unit 3	Functions of random variables of the type $Y=g(X)$, $Z=g(X,Y)$ and two functions of two random variables, Inequalities of Chebyshev and Schwartz. Chernoff Bounds.		
Unit 4	Random processes. First-order stationary processes, second order and wide sense stationary process, Nth order and strict sense stationary process, Time averages and ergodicity, Mean ergodic process, Auto correlation function and its properties, Cross- correlation function and its properties, Covariance functions and their properties, discrete time processes and sequences.		
Unit 5	Power density spectrum and its properties, Linear systems with random inputs. Random signal response, Autocorrelation functions of the response, Cross correlation functions of input and output system, Power density spectrum of the response, Spectral representation. Wiener-Khinchine theorem. Gaussian process, White process, Response of LTI systems to White Noise		
Suggested Readings:			
1. H. Stark and J. W. Woods, Probability and Random Processes with Applications to Signal Processing, 3 rd Edition Prentice Hall.			
2. A Papoulis and S. Pillai, Probability, Random Variables and Stochastic Processes, 4 th Edition, PHI, 2002.			
3. P.Z. Peebles, Probability, Random Variables and Random Signals, 4 th Edition, TMH, 2001.			

C. THEORY LECTURE PLAN

UNIT Nos.		Lecture Nos.	Tut. Classes
	Types of Phenomenon, Theory of Probability	1.	2
	Random experiment, outcome of event, sigma fields	2.	
	The meaning of Probability, Axiomatic Definition of Probability, Probability Operator	3.	
	Conditional Probability & Bayes Theorem	4.	
	Joint events, Joint probability. Repeated Trials. Independence of events Total Probability Theorem,	5.	
	Concept of Random Variable,	6.	2
	Cumulative distribution function and its properties	7.	
	PDF and its properties	8.	
	Probability mass function Discrete Random Variable (Bernoulli, Binomial, Poisson)	9.	
	Gaussian PDF, Gaussian Q-Function, Exponential PDF	10.	
	Mean and Variance of a Continuous and Discrete random Variables	11.	
	Second order Moments	12.	
	Moment Generating Function, Characteristic function	13.	
II.	Concept of Joint Events and Joint Probability	14.	3
	Joint PDF, Joint CDF and their Properties, Independent Random Variables, Conditional Distribution and density function	15.	
		16.	
	Multi-dimensional Gaussian Law	17.	
	Means & Moments, Joint Moments Covariance, Correlation and their properties, Conditional Expectation	18.	
	Joint Moment Generating/Characteristic Function & Problems	19.	
III.	Chebyshev Inequality, Markoff Inequality, Chernoff Bound	20.	2
	Transformation of Random Variables $Y=g(X)$	21.	
		22.	

	Transformation of Random Variables $Z=g(X,Y)$ type	23.	
	Central Limit Theorem, Application of Central Limit Theorem	24.	
IV.	Introduction to Stochastic Processes	25.	
	Mean, Autocorrelation/Autocovariance functions and their properties	26.	
		27.	2
	First-order stationary processes, second order and wide sense stationary process, Nth order and strict sense stationary process,	28.	
		29.	
	Time averages and ergodicity, Mean ergodic process	30.	
	Gaussian Process, White Process, AWGN	31.	
		32.	
	Discrete Time Random Processes: Binomial Counting Process, Random Walk Process	33.	
		34.	
V.	Power density spectrum and its properties	35.	
	Linear systems with random inputs. Random signal response, Autocorrelation functions of the response, Cross correlation functions of input and output system, Power density spectrum of the response, Spectral representation.	36.	
		37.	2
		38.	
	White process, Response of LTI systems to White Noise. Ideal low pass filtered white noise. RC low pass filtered white noise	39.	
		40.	
TOTAL		40	13
TOTAL CLASSES: 53 (Theory + Tutorial)			